Completion-Sector Split Model Note

Internal technical note

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Abstract

This wrapper compiles the split-model section containing the completion-sector assumptions, leak-law organization, threshold analysis, and the theorem-level nondegeneracy clause for the normalized seam coefficient. It serves as a stable ‘xr‘ target for those labels.

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1 Completion-sector leak model and split criterion

This section separates what is directly derived from UOM from the additional hypotheses required to turn those ingredients into a concrete split-driving leak model. The new organizing principle is: the primary hidden variable is a *signed odd split amplitude* a_k, while the positive stock $S_k := a_k^2$ is the quantity that can enter scalar observables and leak laws. We reserve R_k^{abs} for the remaining curvature budget and

$$\mathfrak{r}_k := \min_{\Lambda \geq \Lambda_k} f_k(\Lambda)$$

for the minimized one-wall residual.

1.1 Odd split mode, output parity, and the rank-one normal form

The only “leak” that directly competes with wall work in the Λ-solver is the open-system modification of the wall energy balance allowed explicitly in UOM’s SK/HJ wall equation:

$$\Delta E_{\mathrm{geom},k} \;=\; J_{\mathrm{wall},k} \;+\; Q_{\mathrm{leak},k}, \qquad Q_{\mathrm{leak},k} \geq 0, \tag{1}$$

This is the welded wall-balance form used in the earlier summability analysis. This is a *receiver-side energy complement* term. It must not be conflated with UOM’s information-theoretic bridge/leak instrument from the UOM main manuscript.

What is already derived in UOM. Four ingredients are fixed by the manuscript:

1. **Wall balance.** The reduced one-wall description may contain a nonnegative complement Q_{leak} as in (1).

2. **Quadratic welded SK data.** In the Gaussian sector,

$$W_\Lambda[J] = \frac{1}{2}\int J\,(G_{R,\Lambda} + i\,\mathcal{N}_\Lambda)\,J + O(J^3), \qquad \mathcal{N}_\Lambda \geq 0.$$

3. **Healthy receiver.** The active KR receiver has vanishing welded RP block, so a receiver-tangent RP penalty is not the fundamental home of split-driving oddness.

4. **Continued RP defect.** The analytic-continued covariance has a negative OS mode with

$$\lambda_{\min}(\mu_{ps}, \Lambda) = -\alpha(\Lambda)\mu_{ps}^2 + O(\mu_{ps}^4), \qquad \alpha(\Lambda) \sim \kappa_d\,\Lambda^{d+2}H^{-d}.$$

Assumption 1.1 (Split-mode observability)**.** Near split onset there exists a one-dimensional odd line L_k with local coordinate $a_k \in \mathbb{R}$ such that the split involution acts by

$$a_k \longmapsto -a_k.$$

The true two-wall advantage

$$\Delta F_k := \Delta\mathcal{F}_{2\text{w}-1\text{w},k}$$

and the minimized one-wall residual

$$\mathfrak{r}_k := \min_{\Lambda \geq \Lambda_k} f_k(\Lambda)$$

are smooth even scalar outputs of this same local mode. Equivalently, the 2T Γ-odd descendant direction and the UOM continued negative mode $u_-(\Lambda_k)$ are assumed to represent the same active split mode up to a smooth local identification.

Lemma 1.1 (Evenness of the minimized residual)**.** *Suppose the pre-minimized one-wall residual is a smooth function $f_k(\Lambda, a)$ satisfying*

$$f_k(\Lambda, -a) = f_k(\Lambda, a),$$

and suppose there is a unique local minimizer $\Lambda_*(a)$ near $a = 0$ such that

$$\partial_\Lambda f_k(\Lambda_*(a), a) = 0, \qquad \partial_\Lambda^2 f_k(\Lambda_*(0), 0) > 0.$$

Then

$$\mathfrak{r}_k(a) := f_k(\Lambda_*(a), a)$$

is a smooth even function of a.

Proof. By the implicit function theorem, $\Lambda_*(a)$ is smooth near $a = 0$. Since $f_k(\Lambda, a)$ is even in a, the stationarity equation for a and $-a$ is identical; uniqueness of the local minimizer gives

$$\Lambda_*(-a) = \Lambda_*(a).$$

Therefore

$$\mathfrak{r}_k(-a) = f_k(\Lambda_*(-a), -a) = f_k(\Lambda_*(a), a) = \mathfrak{r}_k(a),$$

so $\mathfrak{r}_k$ is smooth and even. □

Proposition 1.2 (Rank-one normal form near split onset)**.** *Under assumption 1.1, the two scalar outputs admit local even expansions*

$$\Delta F_k(a) = \mu_k^{(2\mathrm{w})} + h_k a^2 + O(a^4), \tag{2}$$

$$\mathfrak{r}_k(a) = \mu_k^{(1\mathrm{w})} + m_k a^2 + O(a^4), \tag{3}$$

with $h_k, m_k > 0$ on the active split line.

Proof. Smooth even functions of one real variable have Taylor series containing only even powers. The constants $\mu_k^{(2\mathrm{w})}$ and $\mu_k^{(1\mathrm{w})}$ are the baseline detunings of the two-wall and one-wall channels. Positivity of h_k and m_k expresses that the selected mode is active in both outputs. □

Corollary 1.3 (Onset wall-match in rank one)**.** *On the onset manifold,*

$$\mu_k^{(1\mathrm{w})} = \mu_k^{(2\mathrm{w})} = 0,$$

so

$$\Delta F_k = h_k a_k^2 + O(a_k^4), \qquad \mathfrak{r}_k = m_k a_k^2 + O(a_k^4).$$

Eliminating a_k^2 gives

$$\Delta F_k = \frac{h_k}{m_k}\,\mathfrak{r}_k + O(\mathfrak{r}_k^2). \tag{4}$$

1.2 Low-rank generalization

If the active split sector is m-dimensional, with odd amplitude vector $a_k \in \mathbb{R}^m$, parity gives quadratic normal forms

$$\Delta F_k(a) = \mu_k^{(2\mathrm{w})} + a^\top H_k a + O(\|a\|^4), \tag{5}$$

$$\mathfrak{r}_k(a) = \mu_k^{(1\mathrm{w})} + a^\top M_k a + O(\|a\|^4), \tag{6}$$

with $H_k, M_k \succeq 0$ on the active split subspace.

Assumption 1.2 (Low-rank comparability)**.** On the active split subspace there exist $c_k^-, c_k^+ > 0$ such that

$$c_k^- M_k \preceq H_k \preceq c_k^+ M_k.$$

Proposition 1.4 (Low-rank wall-match comparability)**.** *Assume assumptions 1.1 and 1.2 and restrict to the onset manifold $\mu_k^{(1\mathrm{w})} = \mu_k^{(2\mathrm{w})} = 0$. Then for sufficiently small $\mathfrak{r}_k$ there exists $C_k > 0$ such that*

$$c_k^-\,\mathfrak{r}_k - C_k \mathfrak{r}_k^2 \le \Delta F_k \le c_k^+\,\mathfrak{r}_k + C_k \mathfrak{r}_k^2. \tag{7}$$

Proof. On onset,

$$\Delta F_k = a_k^\top H_k a_k + O(\|a_k\|^4), \qquad \mathfrak{r}_k = a_k^\top M_k a_k + O(\|a_k\|^4).$$

The quadratic-form comparison gives

$$c_k^- a_k^\top M_k a_k \le a_k^\top H_k a_k \le c_k^+ a_k^\top M_k a_k.$$

Since $\mathfrak{r}_k$ is itself quadratic in a_k to leading order, the quartic remainders are $O(\mathfrak{r}_k^2)$, which yields (7). □

1.3 Gaussian realization of the odd mode and the leak stock law

The theorem path above does *not* rely on a KMS-subtracted matrix. In particular, we do not use

$$\Sigma_{B,k} := \left(C_{B,k} - C^{\mathrm{KMS}}_{\Lambda_k}\right) + m_{B,k} m_{B,k}^{\top}$$

as a positive object, because it need not be positive semidefinite. When a safe matrix observable is needed in the realization layer, the natural PSD quantity is

$$M_{B,k} := C_{B,k} + m_{B,k} m_{B,k}^{\top} \succeq 0. \tag{8}$$

Assumption 1.3 (Gaussian completion realization)**.** For each admissible cutoff Λ in the welded regime, the quadratic influence functional $W_\Lambda[J]$ admits a Gaussian dilation, unique up to symplectic equivalence, by a bosonic completion sector B_Λ with canonical quadratures X_Λ and positive quadratic Hamiltonian

$$H_{B,\Lambda} = \frac{1}{2} X_\Lambda^{\top} h_\Lambda X_\Lambda, \qquad h_\Lambda \succeq 0.$$

Assumption 1.4 (Dominant odd-mode projection)**.** Inside the Gaussian completion of assumption 1.3, there exists a smooth unit vector

$$u_k := u_-(\Lambda_k)$$

spanning the active odd line and separated from the orthogonal sector by a local spectral gap. We write

$$P_k^{\perp} := I - u_k u_k^{\top}$$

for the orthogonal projector.

Assumption 1.5 (Classical-control gate)**.** There exists a scalar gate $g_{\mathrm{cl}}(k) \in [0,1]$, determined by the accessible classicality defect cl_k or Off_k, such that $g_{\mathrm{cl}}(k) \ll 1$ at a fresh-aeon start and $g_{\mathrm{cl}}(k) \to 1$ late in the aeon.

Proposition 1.5 (Projected Gaussian affine dynamics)**.** *Under assumptions 1.3 and 1.4, one may choose a finite effective completion-mode basis in which the completion quadratures obey a linear-Gaussian recursion*

$$X_{k+1} = A_k X_k + B_k j_k^{\mathrm{ctl}} + \nu_k, \tag{9}$$

with

$$\mathbb{E}[\nu_k \mid \mathcal{F}_k] = 0, \qquad \mathbb{E}[\nu_k \nu_k^{\top} \mid \mathcal{F}_k] = Q_k \succeq 0.$$

Consequently the PSD second moment

$$M_{B,k} = C_{B,k} + m_{B,k} m_{B,k}^{\top}$$

obeys

$$M_{B,k+1} = A_k M_{B,k} A_k^{\top} + Q_k + \textit{drift terms}. \tag{10}$$

Projecting onto the odd line via

$$a_k := u_k^{\top} X_k$$

gives the scalar affine law

$$a_{k+1} = \rho_k a_k + \beta_k + \xi_k + \varepsilon_k^{\mathrm{mix}}, \tag{11}$$

where

$$\rho_k := u_{k+1}^{\top} A_k u_k, \qquad \beta_k := u_{k+1}^{\top} B_k j_k^{\mathrm{ctl}},$$

$$\xi_k := u_{k+1}^\top \nu_k, \qquad \varepsilon_k^{\mathrm{mix}} := u_{k+1}^\top A_k P_k^\perp X_k.$$

Moreover,

$$\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0, \qquad \mathbb{E}[\xi_k^2 \mid \mathcal{F}_k] = u_{k+1}^\top Q_k u_{k+1} \geq 0.$$

If assumption 1.5 also holds and the welded shape family is fixed, then

$$\mathbb{E}[\xi_k^2 \mid \mathcal{F}_k] = \eta_k \, g_{\mathrm{cl}}(k) \, \Delta I_k + O(\Delta I_k^2, \delta\Lambda_k^2), \tag{12}$$

with $\eta_k \geq 0$.

Proof. The quadratic SK influence functional determines a Gaussian CPTP map on the finite active completion-mode sector, hence a linear-Gaussian state-space realization of the form (9); (10) is the standard update for the PSD second moment. Decompose $X_k = u_k a_k + P_k^\perp X_k$, apply $\langle u_{k+1}, \cdot \rangle$ to (9), and collect the projected transport, deterministic drive, noise, and orthogonal-mode contamination to obtain (11). The mean-zero and positive-variance statements for ξ_k follow directly from the projected Gaussian noise covariance $u_{k+1}^\top Q_k u_{k+1} \geq 0$. Finally, at fixed welded shape family one has $A_{\mathrm{eff},k}^2 \propto \Delta I_k$, so the fresh covariance injection into the odd mode is linear in the current payload at leading order; the gate $g_{\mathrm{cl}}(k)$ suppresses this injection near a fresh-aeon start and yields (12). □

Assumption 1.6 (Centered odd preparation)**.** Near split onset, the active odd mode is prepared either branch-symmetrically or in a squeeze-dominated regime, so the projected deterministic displacement vanishes and orthogonal-mode contamination is higher order:

$$\beta_k = 0, \qquad \varepsilon_k^{\mathrm{mix}} = O_2.$$

Corollary 1.6 (Centered odd-mode law)**.** *Under assumptions 1.3 to 1.6, the projected affine law* (11) *reduces to*

$$a_{k+1} = \rho_k a_k + \xi_k + O_2, \tag{13}$$

with $\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ *and variance scaling* (12).

Stochastic/Gaussian realization layer. At theorem level the positive stock is simply

$$S_k := a_k^2.$$

The general projected Gaussian dynamics need not be centered. Under assumption 1.6, however, the physically relevant late-time regime is the one in which the odd mode carries no leading deterministic displacement and the fresh effect enters through its variance. In that centered specialization it is natural to use the conditional second moment

$$S_k := \mathbb{E}[a_k^2 \mid \mathcal{F}_k].$$

Proposition 1.7 (Odd-mode stock law)**.** *Under assumptions 1.3 to 1.6, the conditional stock*

$$S_k := \mathbb{E}[a_k^2 \mid \mathcal{F}_k]$$

satisfies

$$S_{k+1} = t_k \, S_k + \eta_k \, g_{\mathrm{cl}}(k) \, \Delta I_k + O(S_k \Delta I_k, \Delta I_k^2, \delta\Lambda_k^2), \qquad t_k = \rho_k^2. \tag{14}$$

Proof. Square (13):

$$a_{k+1}^2 = \rho_k^2 a_k^2 + 2\rho_k a_k \xi_k + \xi_k^2 + O_2.$$

Taking conditional expectation and using $\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ removes the mixed term. Substituting (12) gives (14). □

Assumption 1.7 (UV comparability of energetic stiffness)**.** There exist constants $c_-, c_+ > 0$ and a monotone increasing energetic stiffness $\widetilde{\alpha}(\Lambda)$ of the active odd mode such that, for all sufficiently large Λ,

$$c_- \, \alpha(\Lambda) \le \widetilde{\alpha}(\Lambda) \le c_+ \, \alpha(\Lambda). \tag{15}$$

This is the precise place where the note passes from what is derived in UOM to a physical hypothesis. The coefficient $\alpha(\Lambda)$ is derived from the continued OS problem; the statement that the energetic stiffness of the split mode is comparable to it is not yet a theorem.

With the stock S_k as the actual state variable, the positive split-driving export flux is modeled by

$$Q^{\text{model}}_{\text{leak},k} = \widetilde{\alpha}'(\Lambda_k) \, \delta\Lambda_k \, S_k \; + \; \widetilde{\alpha}(\Lambda_k) \, \eta_k \, g_{\text{cl}}(k) \, \Delta I_k \; + \; O_2, \tag{16}$$

where $\delta\Lambda_k := \Lambda_{k+1} - \Lambda_k$. The first term is the transport cost of carrying already-prepared odd stock to larger cutoff; the second is the fresh injection due to the current payload. Under assumption 1.7, this gives the scaling comparison

$$Q^{\text{model}}_{\text{leak},k} \asymp \alpha'(\Lambda_k) \, \delta\Lambda_k \, S_k \; + \; \alpha(\Lambda_k) \, \eta_k \, g_{\text{cl}}(k) \, \Delta I_k. \tag{17}$$

Current RP proxy reinterpreted. The quantity

$$Q_{\text{rg}} = \chi J_{\text{RP}} / \beta_{\text{dS}}$$

therefore remains useful only as a phenomenological probe. It may correlate with a_k^2 or with the stock S_k, but it is not the fundamental leak law because the healthy receiver itself has no quadratic RP block on the welded family.

1.4 Approximate split condition from the odd-stock leak law

Let $R_k^{\text{abs}} := |R_k|$. In the micro-step regime the geometry-side surrogate gives

$$\Delta E_{\text{geom},k} \approx \frac{\zeta \, (A + c_\Lambda \Lambda_k^2)}{2\sqrt{12 \, R_k^{\text{abs}}}} \, \Theta_k \; - \; \frac{2\zeta \, c_\Lambda \Lambda_k \sqrt{R_k^{\text{abs}}}}{\sqrt{12}} \, \delta\Lambda,$$

while

$$J_{\text{wall},k} \approx \mu_{\text{RG}}(\Lambda_k) \, \delta\Lambda, \qquad \mu_{\text{RG}}(\Lambda) \simeq \mu_0 + \mu_2 \Lambda^2 > 0.$$

Hence the local one-wall threshold remains

$$Q_{\text{leak},k} \gtrsim \frac{\zeta \, (A + c_\Lambda \Lambda_k^2)}{2\sqrt{12 \, R_k^{\text{abs}}}} \, \Theta_k. \tag{18}$$

Since

$$\Theta_k = K(\Lambda_k) \, \Delta I_k \sim K_0 \, \Lambda_k^{-p} \, e^{-2 d_{\text{sep}}(\Lambda_k) \Lambda_k} \, \Delta I_k, \qquad p \in \{10, 14\},$$

substituting (16) gives

$$\widetilde{\alpha}'(\Lambda_k) \, \delta\Lambda_k \, S_k \; + \; \widetilde{\alpha}(\Lambda_k) \, \eta_k \, g_{\text{cl}}(k) \, \Delta I_k \gtrsim \frac{\zeta \, (A + c_\Lambda \Lambda_k^2)}{2\sqrt{12 \, R_k^{\text{abs}}}} \, K(\Lambda_k) \, \Delta I_k. \tag{19}$$

This is the corrected split threshold. It shows the two late-time branches transparently:

1. if S_k remains negligible, only the fresh term remains, and the exhaustion branch can survive;
2. if S_k has accumulated, the transport term can dominate even when ΔI_k is already tiny.

Under assumption 1.7 and $\alpha(\Lambda) \sim \kappa_d \Lambda^{d+2} H^{-d}$, the transport term scales like

$$\widetilde{\alpha}'(\Lambda_k)\, \delta\Lambda_k\, S_k \asymp \Lambda_k^{d+1}\, \delta\Lambda_k\, S_k,$$

which is exactly the mechanism by which late split can beat the shrinking fresh payload.

1.5 Nondegeneracy clause for the normalized seam coefficient

The split threshold (19) is a statement about when the late-time odd stock can overcome the shrinking fresh payload. It is *not* by itself a theorem that the normalized seam coefficient c_{Λ_k} on the canonical compressed welded LoG line is nonzero. Any theorem proving $c_{\Lambda_k} \neq 0$ must therefore come from a separate overlap or transport hypothesis, not from the threshold inequality alone. Downstream, this clause is absorbed into the delayed selected-line package rather than carried as a separate package summand.

Definition 1.8 (Unit-source seam overlap scalar and no-destructive-cancellation margin)**.** Fix an accepted tick k. Assume the canonical compressed welded LoG line is defined and let

$$\tau_{\Lambda_k}^{\mathrm{unit}}$$

be a detector-factorized unit welded odd source. Define the raw unit-source seam overlap scalar by

$$\widetilde{c}_{\Lambda_k} := \left\langle\!\!\left\langle \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{\mathrm{unit}}\big],\, \delta C_{\Lambda_k}^{\mathrm{LoG}} \right\rangle\!\!\right\rangle_{C_{*,\Lambda_k}}. \tag{20}$$

A *no-destructive-cancellation margin* at Λ_k is any real number

$$\mu_{\Lambda_k} > 0$$

such that

$$|\widetilde{c}_{\Lambda_k}| \geq \mu_{\Lambda_k}. \tag{21}$$

On a connected accepted-branch segment I, a *branchwise no-destructive-cancellation margin* is a constant $\mu_I > 0$ such that

$$|\widetilde{c}_{\Lambda}| \geq \mu_I \qquad \forall\, \Lambda \in I. \tag{22}$$

Theorem 1.9 (Nondegeneracy criterion for c_{Λ_k})**.** *Fix an accepted tick k. Assume:*

1. *a detector-factorized unit welded odd source at Λ_k;*
2. *the unit-source seam-faithfulness package, namely:*
   (a) *the raw seam pairing factorizes as*
   $$\widetilde{s}_{\Lambda_k}(f, y) = \sigma_{\mathrm{fill}}(f)\, \widetilde{c}_{\Lambda_k}\, \overline{\mathrm{obs}}(y) \qquad \forall f, \forall y; \tag{23}$$
   (b) *whenever $\widetilde{c}_{\Lambda_k} \neq 0$ and $\overline{\mathrm{obs}}(y) \neq 0$, the compressed welded LoG line is transversely physical, dominant-line realization holds, and the normalized seam coefficient*
   $$c_{\Lambda_k} := \kappa_{\Lambda_k}^{-1/2}\, \widetilde{c}_{\Lambda_k} \tag{24}$$
   is defined;

3. *a no-destructive-cancellation margin at Λ_k in the sense of definition 1.8;*

4. *a witness y with*
$$\overline{\mathrm{obs}}(y) \neq 0.$$

Then:

1. *the raw seam pairing is nonzero for every filler tag f:*
$$\widetilde{s}_{\Lambda_k}(f, y) \neq 0;$$

2. *the canonical compressed welded LoG line is transversely physical at Λ_k;*

3. *dominant-line realization holds at Λ_k;*

4. *the normalized seam coefficient is defined and nonzero:*
$$c_{\Lambda_k} \neq 0.$$

If, in addition, for some filler tag f the source $\tau_{\Lambda_k}^{f,y}$ is representable on the characterization layer and admits a source-compatible descendant realization, then the corresponding descendant representative satisfies
$$\rho_{\Lambda_k}^{\mathrm{dom}}\big(J_{\Lambda_k}^{f,y}\big) \neq 0.$$

Proof. The margin (21) implies $\widetilde{c}_{\Lambda_k} \neq 0$. Together with $\overline{\mathrm{obs}}(y) \neq 0$, the factorization (23) gives $\widetilde{s}_{\Lambda_k}(f, y) \neq 0$ for every filler tag f, proving item (1). Item (2), item (3), and the definition and non-vanishing of c_{Λ_k} are exactly the remaining conclusions of the assumed unit-source seam-faithfulness package under the same nondegeneracy hypothesis. The final sentence is the corresponding witness-to-descendant comparison step under representability and source-compatible descendant realization. □

Corollary 1.10 (Branchwise sign-stable nondegeneracy)**.** *Let I be a connected accepted-branch segment. Assume:*

1. *the branchwise normalized seam coefficient*
$$\Lambda \mapsto c_\Lambda$$
is continuous on I;

2. *a branchwise no-destructive-cancellation margin on I in the sense of definition 1.8.*

Then
$$c_\Lambda \neq 0 \qquad \forall\, \Lambda \in I.$$
In particular, the sign of c_Λ is constant on I. If there also exist a fixed LS witness y with $\overline{\mathrm{obs}}(y) \neq 0$ and a branchwise source-compatible descendant realization satisfying
$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y),$$
then the sign of $\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{f,y})$ is constant on I as well.

Proof. By (22), the scalar $\widetilde{c}_\Lambda$ never vanishes on I. Hence neither does c_Λ, because the normalization only rescales the raw overlap scalar by a positive branchwise factor. A continuous nonzero real-valued function on a connected segment cannot change sign, proving the constancy of $\operatorname{sgn}(c_\Lambda)$. The final statement follows immediately from the displayed branchwise comparison identity and the fixed nonzero scalar $\overline{\mathrm{obs}}(y)$. □

Remark 1.11 (Logical separation from the split threshold). The hypotheses of theorem 1.9 and corollary 1.10 are intentionally separate from (19). The threshold governs the emergence of split-side stock; the no-destructive-cancellation margin governs overlap of the unit welded odd source with the canonical compressed dominant line. Accordingly, the present note treats $c_\Lambda \neq 0$ as the transport/overlap clause later imported into the delayed selected-line package rather than as a consequence of the split threshold.